Novel AI/ML & Computational Science Methods in Pharmaceutical Workflows

Based on articles retrieved from PubMed | Literature searches conducted May 2026 | All DOIs linked

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Overview

Artificial intelligence and machine learning are reshaping every stage of the pharmaceutical pipeline—from how researchers synthesize knowledge, to how molecules are designed, modeled, and monitored after launch. Below is a structured review by workflow stage, grounded in recent literature (2023–2026).

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1. Literature & Background Research

Named Entity Recognition (NER) and Text Mining

The volume of biomedical literature now outpaces any team's ability to read it manually. NLP methods—particularly transformer-based models like BioBERT and PubMedBERT—now automate the extraction of drug–gene, drug–disease, and drug–protein relationships from millions of abstracts.

A 2025 review from EMBL-EBI describes the evolution of NLP in drug discovery from rule-based ontology tagging to ML-driven entity identification, highlighting that methods must be matched to the specific use case and data available, as each approach has distinct performance conditions (Withers et al., 2025).

The DrugProt initiative at BioCreative VII demonstrates scalability: deep learning systems built on BERT/BioBERT applied to over 5,000 PubMed abstracts generated a **silver-standard knowledge graph with ~54 million nodes and ~19 million edges** encoding chemical–protein interactions, with precision and recall values reaching 0.92 and 0.95 for specific relation types (Miranda-Escalada et al., 2023).

SciLinker, a modular NLP pipeline applied to the entire PubMed corpus, identified over **30 million association sentences** and more than 1.25 million unique gene–disease associations, enabling disease-specific network construction for target identification (Liu et al., 2025).

Automated Target Discovery Pipelines

LLM-powered pipelines are now used to screen literature for novel therapeutic targets, particularly in rare diseases where structured databases are sparse. TARGETFLOW, an automated LLM-based mining pipeline, combines abstract retrieval, text cleaning, intelligent screening, and rule-based filtering to surface high-potential targets. Validated on rheumatoid arthritis and idiopathic pulmonary fibrosis, it achieved a **56% high-potential target validation rate** and a **100% novel-target validation pass rate** for targets not yet in existing databases (Guo et al., 2025).

Knowledge Graph Construction

Beyond raw text mining, systems now automatically construct structured knowledge graphs linking genes, diseases, drugs, and pathways. These provide the semantic scaffolding for downstream modeling—connecting experimental findings to regulatory context and mechanistic hypotheses. The PRESS framework extends this by pairing BioBERT-extracted gene-interaction knowledge with S-system differential equation models for gene regulatory network reconstruction, substantially reducing parameter estimation complexity (Gill et al., 2025).

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2. Model Building (Target Identification & Lead Discovery)

AlphaFold and Structural Prediction

DeepMind's AlphaFold 2 (and its successors) fundamentally altered structural biology. By predicting protein structures at near-crystallographic accuracy, it enabled drug discovery against previously intractable targets—those lacking experimental holo structures or known small-molecule binders. A 2025 review documents AlphaFold 3's further breakthroughs in predicting multi-molecular complexes including protein–ligand and protein–nucleic acid interactions, with regulatory validation of digital twin methodologies now underway (Wan et al., 2025).

Contrastive Learning for Genome-Wide Virtual Screening

A landmark 2026 paper in *Science* describes **DrugCLIP**, a contrastive learning framework enabling virtual screening up to **10 million times faster** than traditional molecular docking. Tested on ~10,000 human proteins screened against 500 million compounds, it achieved a 15% hit rate against the norepinephrine transporter and a 17.5% hit rate against a target (TRIP12) using only AlphaFold2-predicted structures. The resulting open-access GenomeScreenDB provides precomputed genome-wide screening results, marking a paradigm shift in post-AlphaFold drug discovery (Jia et al., 2026).

Graph Neural Networks (GNNs) for Drug–Target Interactions

Molecular structures and protein residue networks are naturally represented as graphs—making GNNs a leading architecture for binding affinity and interaction prediction. Key recent advances include:

- **SGADN** (Structure-Aware Graph Attention Diffusion Network): models protein–ligand complexes as line graphs incorporating both distance and angle information, with an attentive pooling layer to capture hierarchical binding structures, outperforming baselines on standard benchmarks (Li et al., 2024).
- **iNGNN-DTI**: an interpretable nested GNN that integrates AlphaFold2-derived 3D target structures with pre-trained small-molecule representations, using cross-attention to reveal substructure-level interaction hotspots (Sun et al., 2024).
- **MM-TCoCPIn**: a multi-modal GNN combining network topology (CTC-based encoder), biomedical text semantics (SciBERT), and 3D protein geometry (GVP-GNN on AlphaFold2 contact graphs), achieving AUC = 0.93 and F1 = 0.92 for chemical–protein interaction prediction with ablation-validated modality contributions (Wang, 2025).
- **Spatom**: a weighted directed-graph convolutional network with improved graph attention for predicting protein–protein interaction sites, relevant to antibody and PPI-targeting drug design (Wu et al., 2023).

A comprehensive review situates GNNs within the broader DTI prediction landscape, cataloguing public databases and benchmarks (Zhang et al., 2022).

Generative Models for De Novo Molecular Design

Deep generative models now synthesize drug-like molecules from scratch rather than screening pre-existing libraries:

- **Variational autoencoders (VAEs)** learn latent chemical space representations and sample novel structures from that space.
- **Generative adversarial networks (GANs)**: DNMG uses a Wasserstein-GAN architecture with 3D grid spatial information and atomic physicochemical properties to generate ligand representations parsed into SMILES strings, demonstrating superior binding affinity for MK14, FNTA, and CDK2 targets (Song et al., 2023).
- **Transformer-based and hybrid models**: recurrent neural networks, autoencoders, and reinforcement learning–augmented transformers are described as enabling exploration of vast chemical space with targeted property optimization (Tong et al., 2021).
- A Yale survey covering both small-molecule and protein generation organizes the field into subtasks (scaffold-based, fragment-based, goal-directed, protein sequence/structure design) and benchmarks leading architectures (Tang et al., 2024).
- A focused review on deep generative models surveys molecule representations, data processing pipelines, and generation frameworks, identifying gaps including limited 3D structural information and the need for higher-precision evaluation metrics (Pang et al., 2023).

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3. Implementation (Preclinical to Clinical)

ADMET and Druggability Prediction

Early prediction of absorption, distribution, metabolism, excretion, and toxicity (ADMET) properties filters out problematic candidates before expensive wet-lab work. ML models—from gradient-boosted trees to deep learning—predict solubility, activity, toxicity, pharmacokinetics, and drug-likeness against ultra-large compound libraries, accelerating lead compound screening (Li et al., 2025).

Ensemble ML approaches (combining up to 101 algorithms) are being used for multi-property predictive models. In one example, ADMET properties including molecular docking scores, LibDock scores, and toxicity reduction were integrated to optimize quercetin derivatives as candidates for chronic kidney disease (Xing et al., 2025).

AI in Pharmacokinetics and Pharmacodynamics

Hybrid and purely ML-based PK/PD models now complement traditional compartmental approaches:

- **NeuralODEs and recurrent architectures** (e.g., LSTM) handle sparse, irregular clinical pharmacokinetic data better than classical methods.
- **Model-informed precision dosing (MIPD)**: population PK/PD models combined with ML allow adaptive, individualized dosing regimens. In precision antibiotic dosing (e.g., linezolid), MIPD using early drug concentration measurements and platelet dynamic modeling provides prospective toxicity prevention (Takahashi & Tsuji, 2026).
- A 2025 review in *Therapie* maps AI's expanding role across PK (exposure prediction, sparse data modeling), PD (precision medicine, adaptive regimens), and pharmacovigilance—while noting limitations in model explainability and training data quality (Gérard et al., 2025).

AI in Clinical Trial Optimization

Persona-driven generative AI frameworks apply contextual LLM "personas" (structured domain-expert roles) across clinical development tasks—from drafting protocols to interpreting regulatory guidance. A 2025 *Drug Discovery Today* review documents 60–70% timeline reductions and improved clinical trial efficiency using these approaches, while flagging ongoing challenges in hallucination mitigation and regulatory validation (Wan et al., 2025).

Broader AI-driven drug development reviews highlight clinical trial optimization, regulatory intelligence, and patient stratification as key emerging applications of NLP and ML (Jarallah et al., 2025).

Pharmaceutical Manufacturing: ML and Digital Twins

In process manufacturing of oral solid dosage forms, ML is being integrated with Quality-by-Design (QbD) and Industry 4.0 frameworks:

- ML models predict **critical quality attributes** (CQAs) such as granule size distribution, moisture content, and tablet hardness from process parameters in real time.
- **Process Analytical Technology (PAT)** with adaptive ML enables real-time quality control, replacing end-of-batch testing.
- Key challenges include data availability across scales, model interpretability, uncertainty quantification, and integration with legacy manufacturing workflows (Rezaeizadeh et al., 2025).

A companion review positions AI alongside IoT, robotics, blockchain, and digital twins as drivers of pharmaceutical Industry 4.0, with personalized medicine (leveraging genomic, clinical, and environmental data) as a key downstream capability (Suriyaamporn et al., 2025).

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4. Output (Safety, Monitoring & Post-Market Surveillance)

AI in Pharmacovigilance

Post-market safety monitoring is one of the most data-intensive areas of pharmaceutical practice. AI has improved pharmacovigilance by:

- **Automated signal detection**: data mining and automated signal detection algorithms (e.g., MGPS, PRR, BCPNN, ROR) now rapidly identify disproportionate adverse drug reaction (ADR) reporting across millions of FAERS records.
- **Bayesian networks**: expert-defined Bayesian network tools for causality assessment have reduced ADR processing times from days to hours at pharmacovigilance centers while reducing inter-rater subjectivity (Algarvio et al., 2025).
- **Duplicate detection** and NLP-driven structured extraction from electronic health records (EHRs) deepen real-world evidence analysis beyond structured databases.
- **Predictive models for ADRs and drug–drug interactions** allow proactive patient care, including early flagging of drug-associated pregnancy risks.

Explainable ML for ADR Risk Stratification

A 2026 study applying XGBoost with SHAP (Shapley Additive exPlanations) to the FAERS database identified immunomodulators, antiviral drugs, and neuropsychiatric agents as top predictors of drug-associated miscarriage. The model achieved AUC = 0.738, and SHAP analysis quantified feature-level contributions (e.g., adalimumab and infliximab ranked as the strongest predictors), providing interpretable evidence for individualized risk assessment by age and weight (Lin et al., 2026).

This exemplifies the broader shift toward **explainable AI (XAI)** in pharmaceutical outputs—where not just prediction accuracy but model transparency is required for regulatory acceptance and clinical trust.

Cross-Cutting Themes and Challenges

Several computational themes cut across all workflow stages:

Multi-modal data integration is increasingly common—combining structural (3D protein geometry), semantic (literature), and network-topological features into unified ML architectures. **Transfer learning and pre-trained molecular/protein foundation models** (e.g., BERT variants, ESM protein language models) reduce the labeled-data burden by leveraging large unsupervised corpora before fine-tuning on specific tasks.

Interpretability and explainability remain key challenges. "Black box" deep learning models create regulatory and clinical trust barriers; SHAP, attention visualization, and cross-attention modules are increasingly built into architectures to address this. **Data quality and standardization**—both in training datasets and real-world evidence—is a limiting factor cited across nearly all reviewed domains.

Finally, **digital twins** represent the convergence of many of these methods: AI/ML models continuously updated with real-time process or patient data to simulate, predict, and optimize—from manufacturing reactors to individual patient dosing.

Reference List (PubMed)

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